Day 20, Unit 2, October 11 & 12

Agenda:

1. HDI Project Feedback
2. 2.3 Population Composition
3. Population Pyramid Analysis

Warm Up:

1. What will happen to Rwanda’s population? Canada’s population? Japan’s population?
2. What happens to the life expectancy and infant mortality as a country develops?
3. How can a population pyramid depict an event in the past?

Today’s Objective

1. Describe elements of population composition used by geographers.
2. Explain ways that geographers depict and analyze population pyramids.

What is a population pyramid?

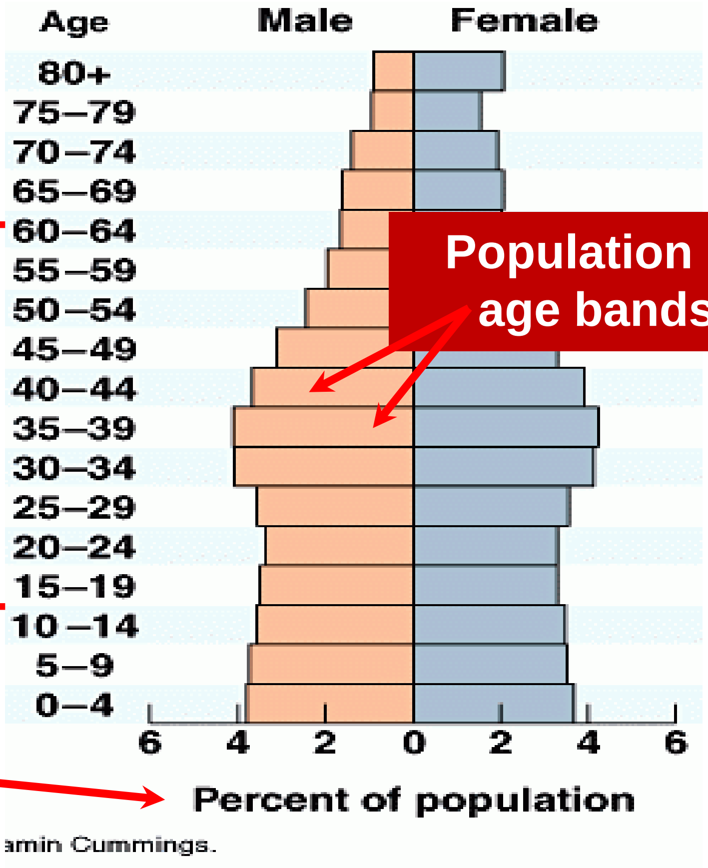
- Graphic visual
- Used to analyze the population growth or decline of a given area
- Shows distribution of various age groups, the sex ratio, and the dependency ratio

Population Pyramids

“The Oldies”

Economically Active

The Kiddos



Population in age bands

This can be a percentage or a #

Population Pyramids gives us an idea about...

- Life Expectancy
- Birth Rate (CBR)
- Death Rate (CDR)
- Literacy Rate
- Infant Mortality Rate (IMR)
- Poverty Level

Would these indicators be high or low for developed and developing countries? Mark in your notes...

Some Math :)

Sex ratio: # of males compared to females

2. Calculation:

$$\frac{\text{Number of Resident Male Live Births}}{\text{Number of Resident Female Live Births}} \times 100 \text{ (or 1,000)}$$

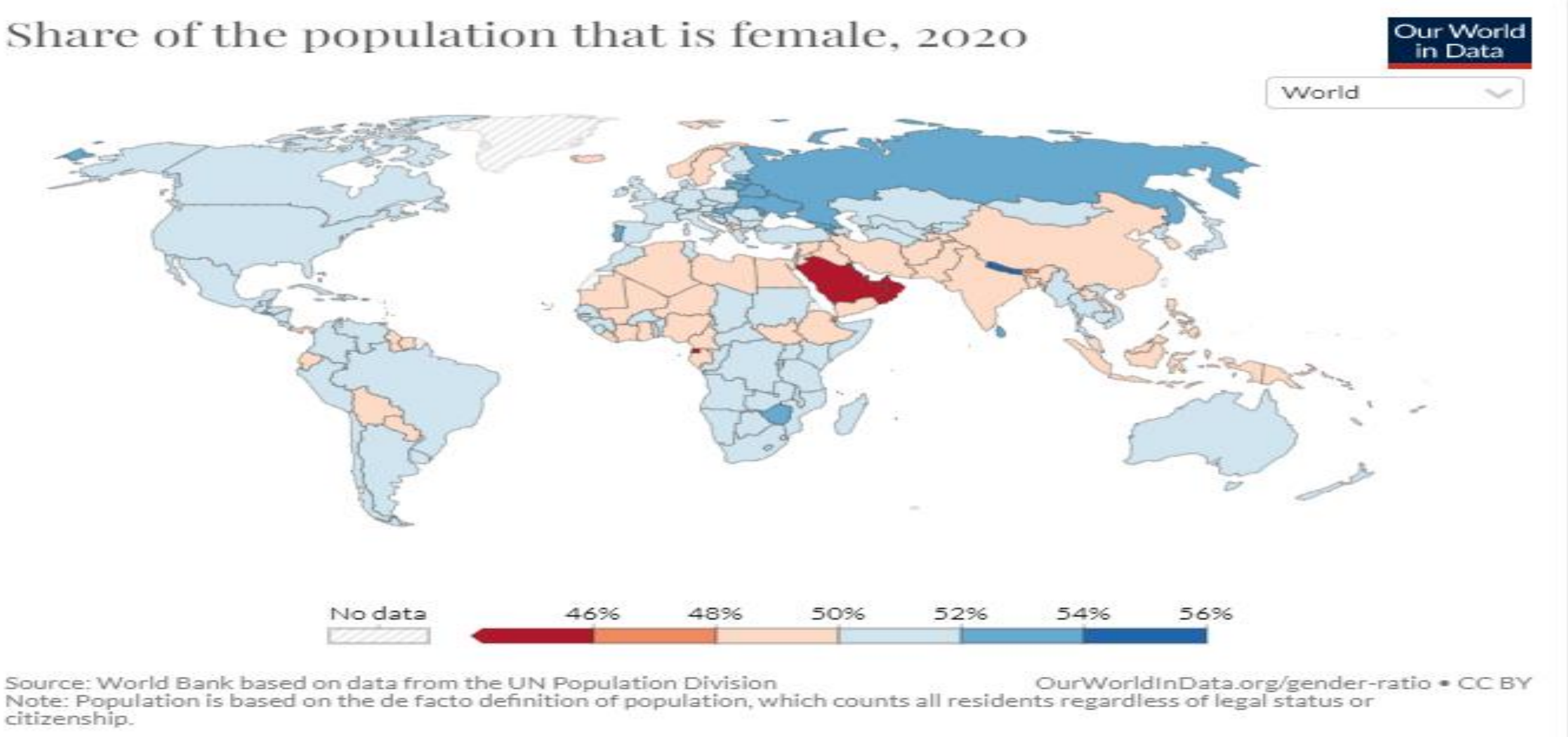
3. Example:

58,000 = male live births in 2008 to state residents
55,000 = female live births in 2008 to state residents
(58,000/55,000) x 100 = 105.5 male births per 100 female live births in 2008 among state residents
(NOTE: If 1,000 is used as the multiplier instead of 100, the result would = 1,055 male births per 1,000 female births.)

1:1 (100:100) is perfectly balanced in theory; in reality due to biology, boys are born more frequently than females

103-107: 100 - is the “natural” balanced sex ratio; Why?

[View of share of females in Population by country]: click on link for better view and trend data



Dependency Ratios

The total dependency ratio is calculated as:

(([Population ages 0-15] + [Population ages 65-plus]) ÷ [Population ages 16-64]) × 100

The old-age dependency ratio is calculated as:

([Population ages 65-plus] ÷ [Population ages 16-64]) × 100

The formula for the youth dependency ratio is:

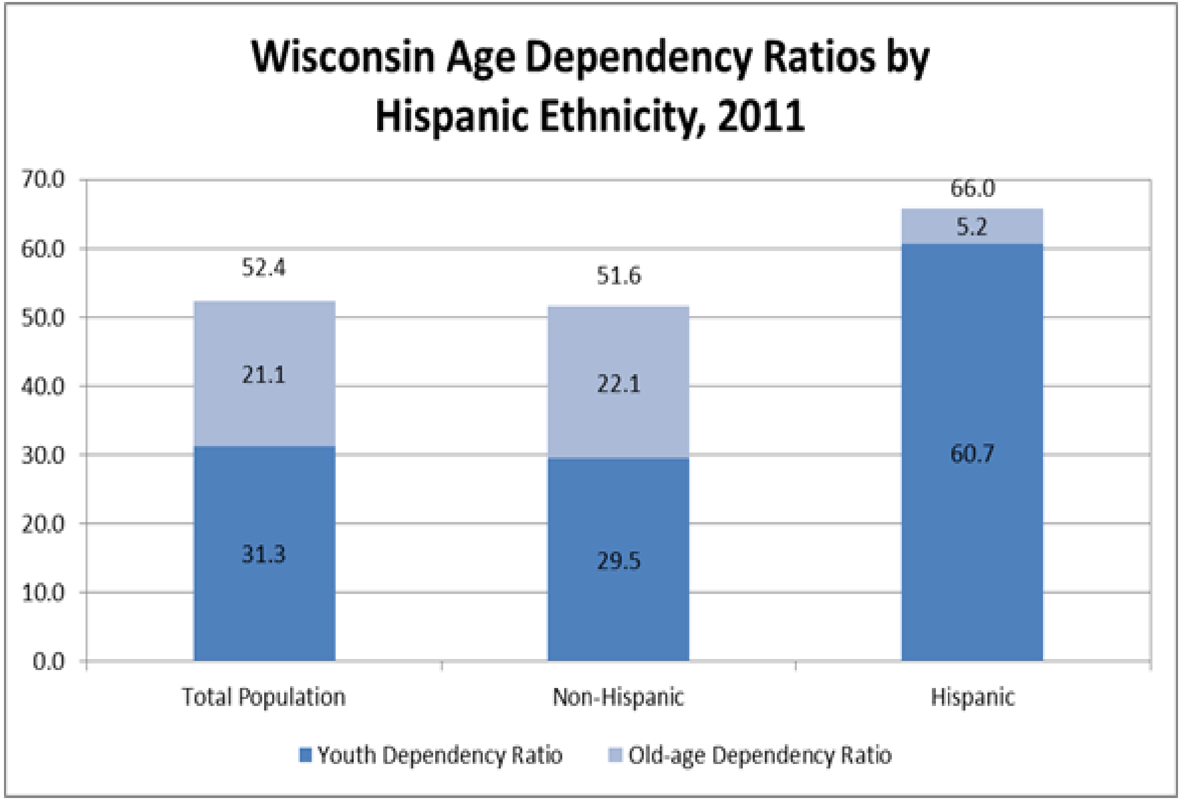
([Population ages 0-15] ÷ [Population ages 16-64]) × 100

It is important to note that these definitions do not take into account labor force participation rates by age group. Some portion of the "dependent" population may be employed and not necessarily economically dependent. The Bureau of Labor Statistics addresses this through a related measure, the economic dependency ratio, which includes survey-based estimates of labor force participation rates by age group.

As an example, the chart below shows dependency ratios for 2011 by Hispanic ethnicity in Wisconsin. The total dependency ratio for non-Hispanics is 51.6, slightly lower than the total of 52.4 for Wisconsin. The total dependency ratio for Hispanics is higher, at 66.0. The youth and old-age dependency ratios differ dramatically between the Hispanic and non-Hispanic populations. The Hispanic youth dependency ratio of 60.7 is much higher than the non-Hispanic ratio of 29.5, while the Hispanic old-age dependency ratio of 5.2 is much lower than the non-Hispanic ratio of 22.1.

These ratios indicate that while the total age dependency in the Hispanic population is higher compared to non-Hispanics, this is entirely due to the relative youthfulness of the Hispanic population.

Example from Wisconsin Government Website



U.S. Dependency Ratio compared to the world

https://www.worldeconomics.com/Demographics/Age-Dependency-Ratio-Total/United%20States.aspx

The total age dependency ratio is the ratio of young + elderly dependents (who are generally economically inactive, under 15 or over 64 years old), compared to the number of people of working age (15-64-year-olds).

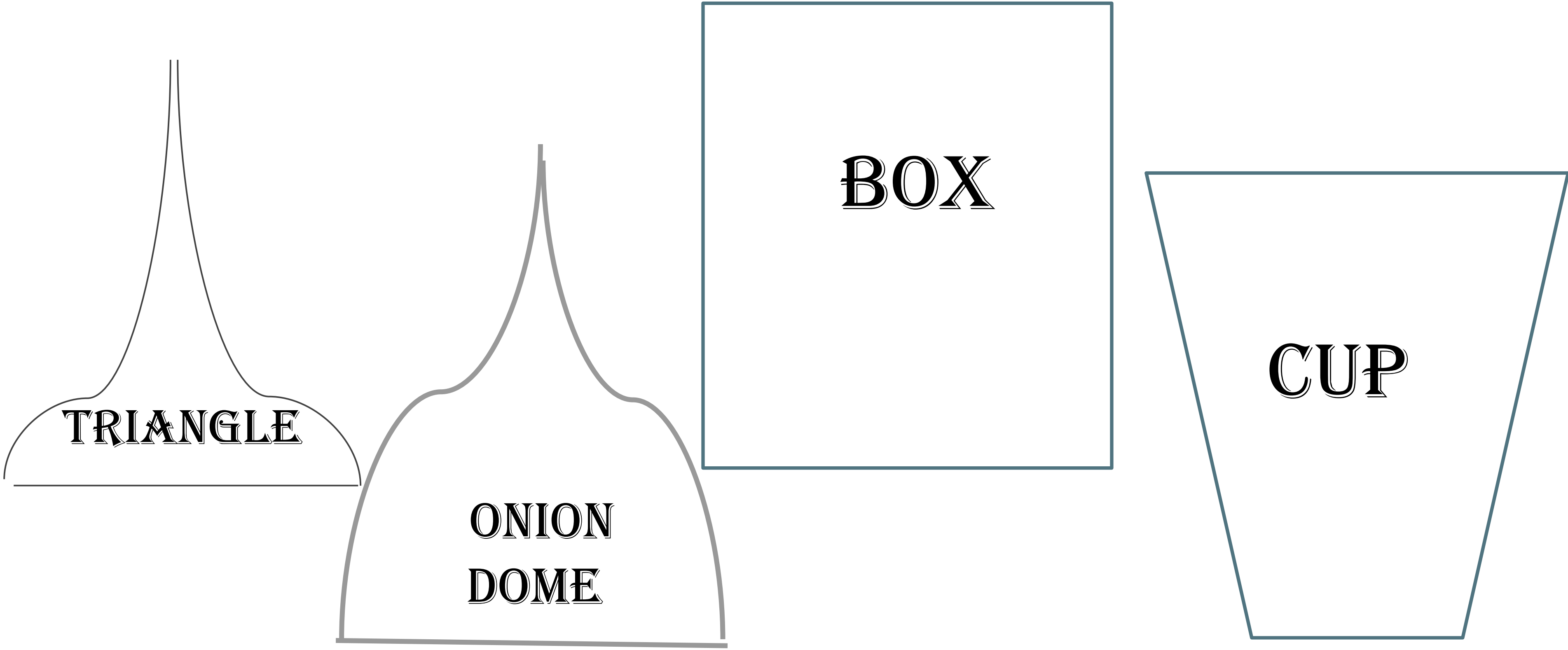
A high dependency ratio means those of working age, and the overall economy, face a greater burden in supporting the dependent population.

United States's age dependency ratio for the dependent population was: 46.1% reported in 2019 (most recent observation). This is a high value against a global average of 40.1%. A higher ratio indicates more financial stress on working people and possible political instability.

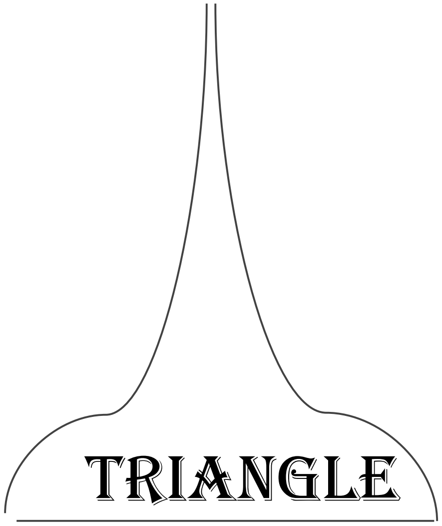
United States's data is highlighted in the table below, use the filter and sort order options to allow easy comparison with other countries.

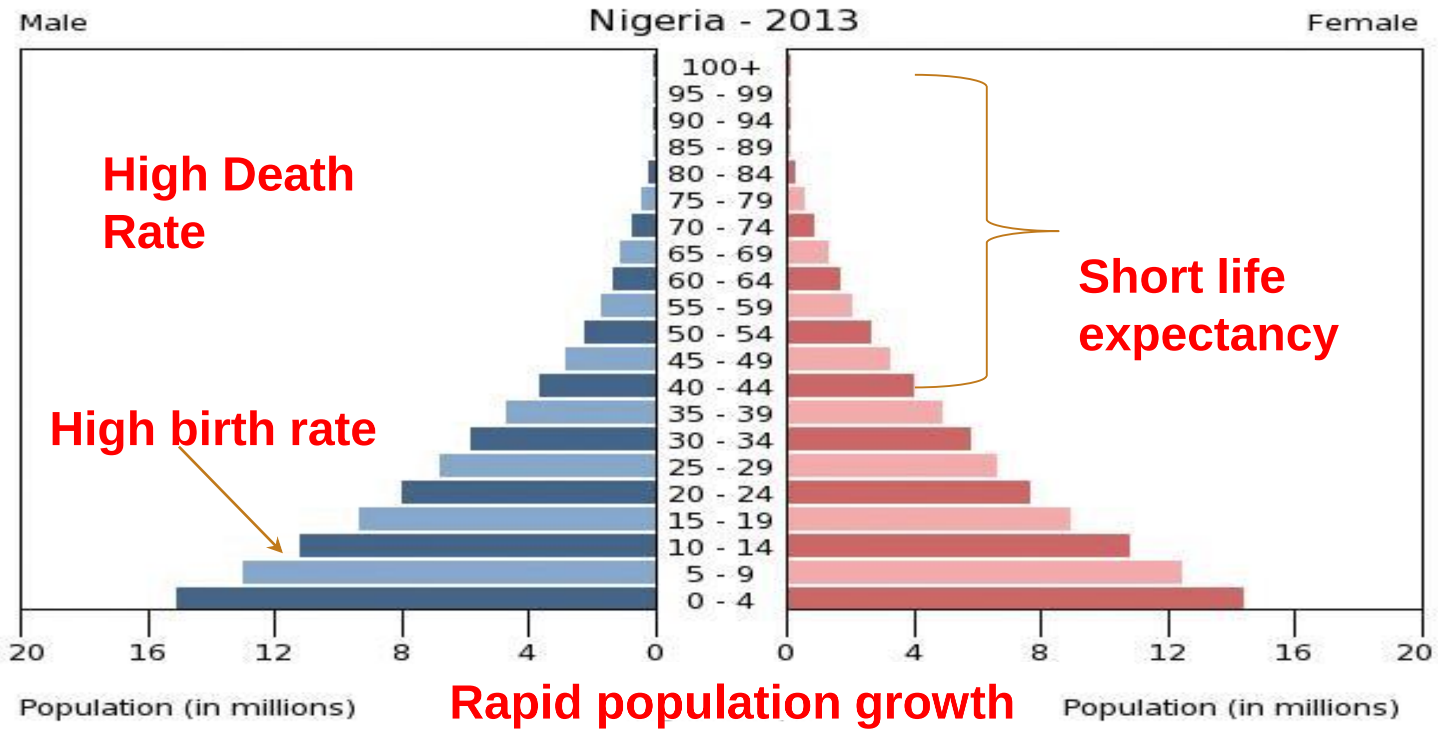
Data source: World Bank, Washington D.C.

Types of Population Pyramids



Triangle = Developing/Periphery





Characteristics

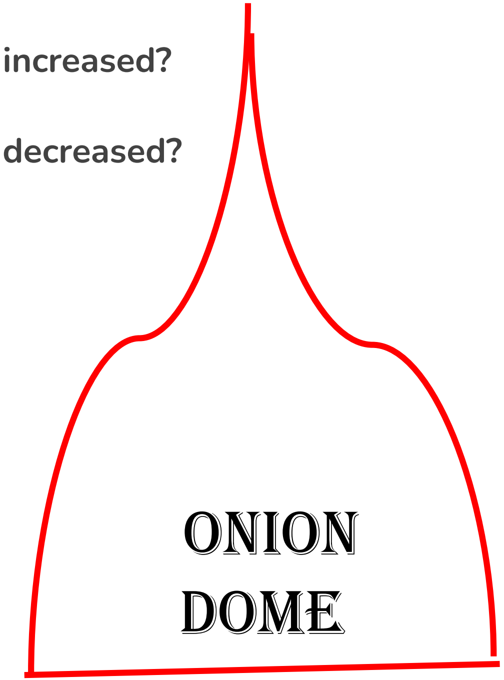
- Rapidly growing pop. → Low resources
  - *(country may exceed carrying capacity!)*
- High CBR, CDR, TFR, and high infant mortality
- Level of development: LDCs
- Most of pop. are kids → Gov't places more resources in child care/benefits (High youth dependency)
- Primarily Primary Sector (like Agriculture) dominant
- Low quality of life, civil strife/political instability/Economic dependency
- EX: Chad

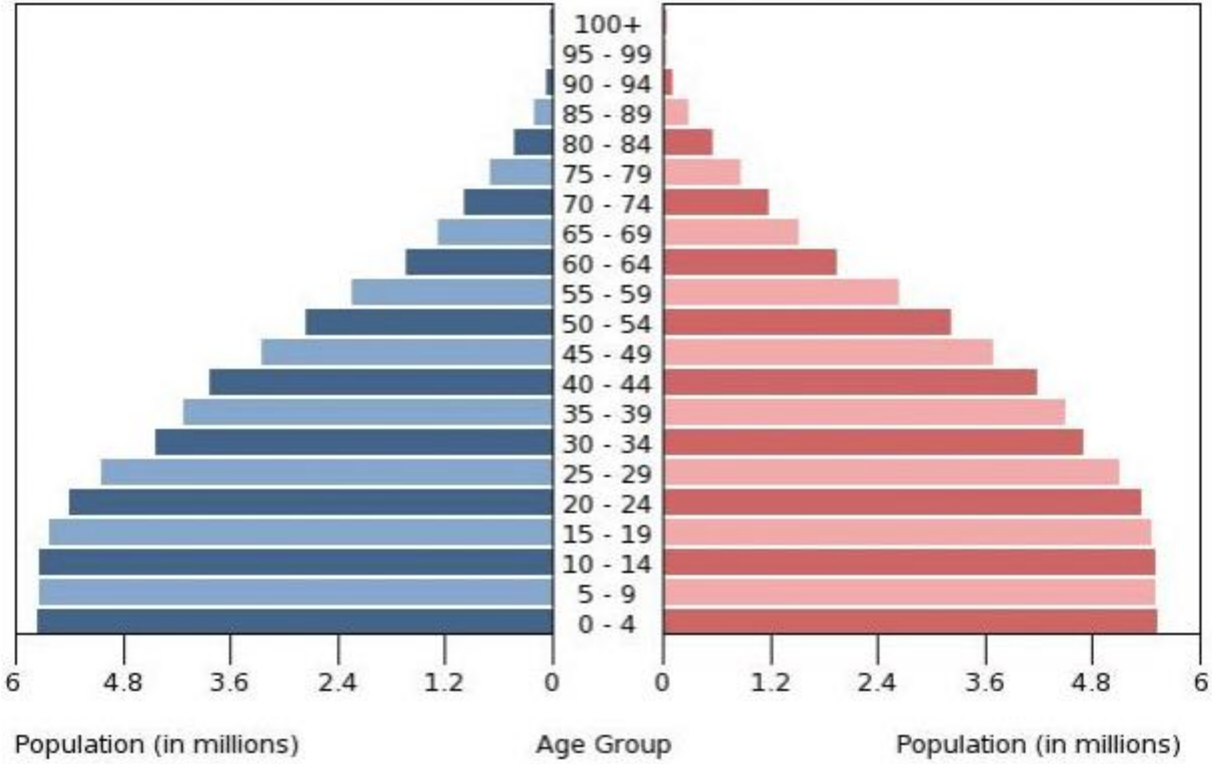
The "Onion Dome" = NIC(Moderate/Slowing Growth)

Newly Industrialized Countries (NICs)

What increased?

What decreased?



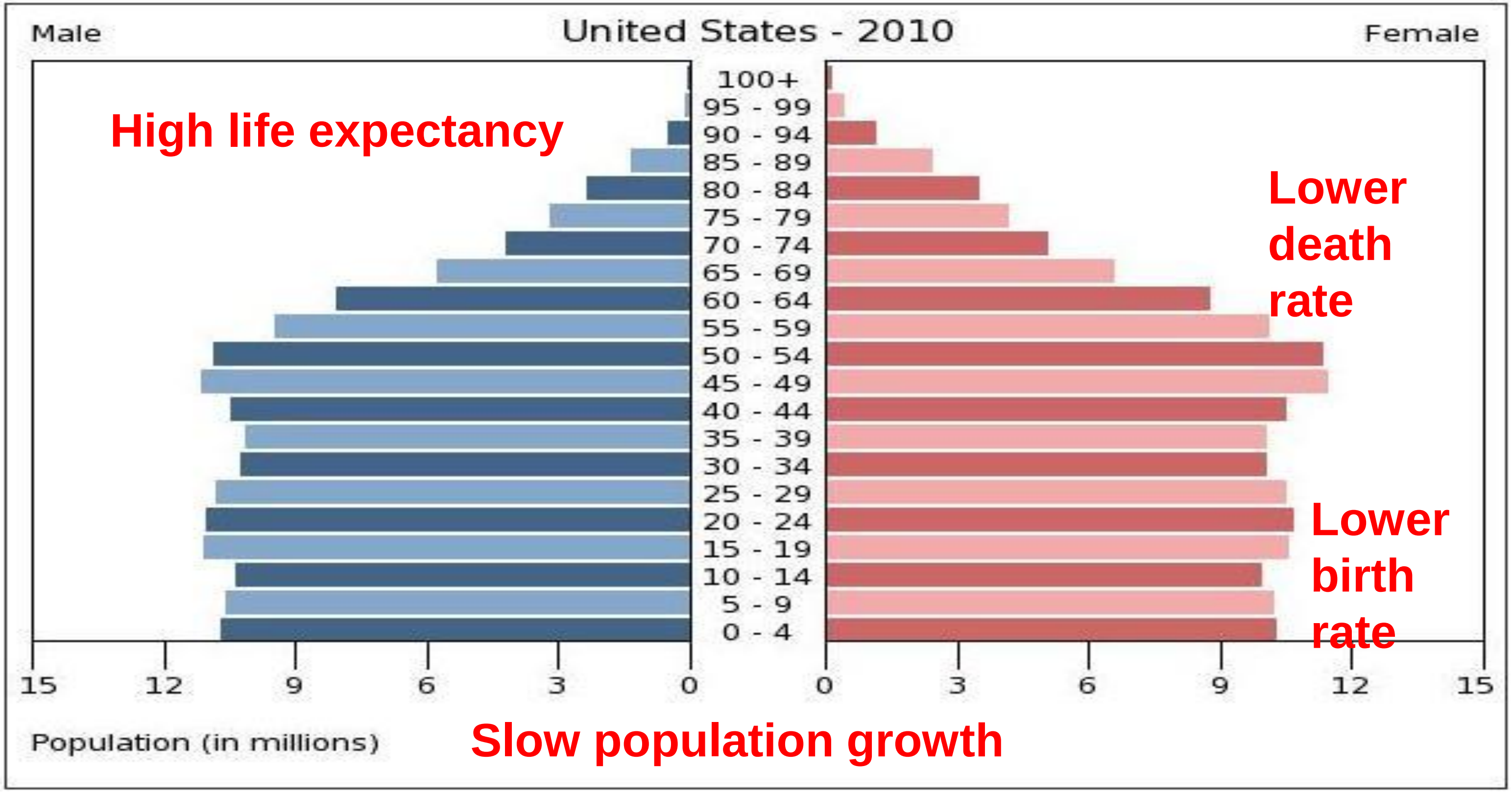


Characteristics

- Moderate/Slow Growth
- CBR, CDR begin to decrease
- Declining IMR, Life expectancy increasing
- TFR may still be relatively high
- Development: NICs or LDCs in transition
- Youth dependency ratio while high is stabilizing, Elderly ratio will begin to rise
- Focus on increasing education quality and access/opportunities
- May still be concerned about continued population growth; reaching carrying capacity; may seek lower TFR via govt. Programs as pop still has large numbers of youths
- Moving away from dependency on foreign aid; working to diversity economic activities, workforce
    - Secondary/Industry is increasing or high

Example: India

Box = Developed/Core (Very Slow or Zero Growth)



THE BOX!

Characteristics

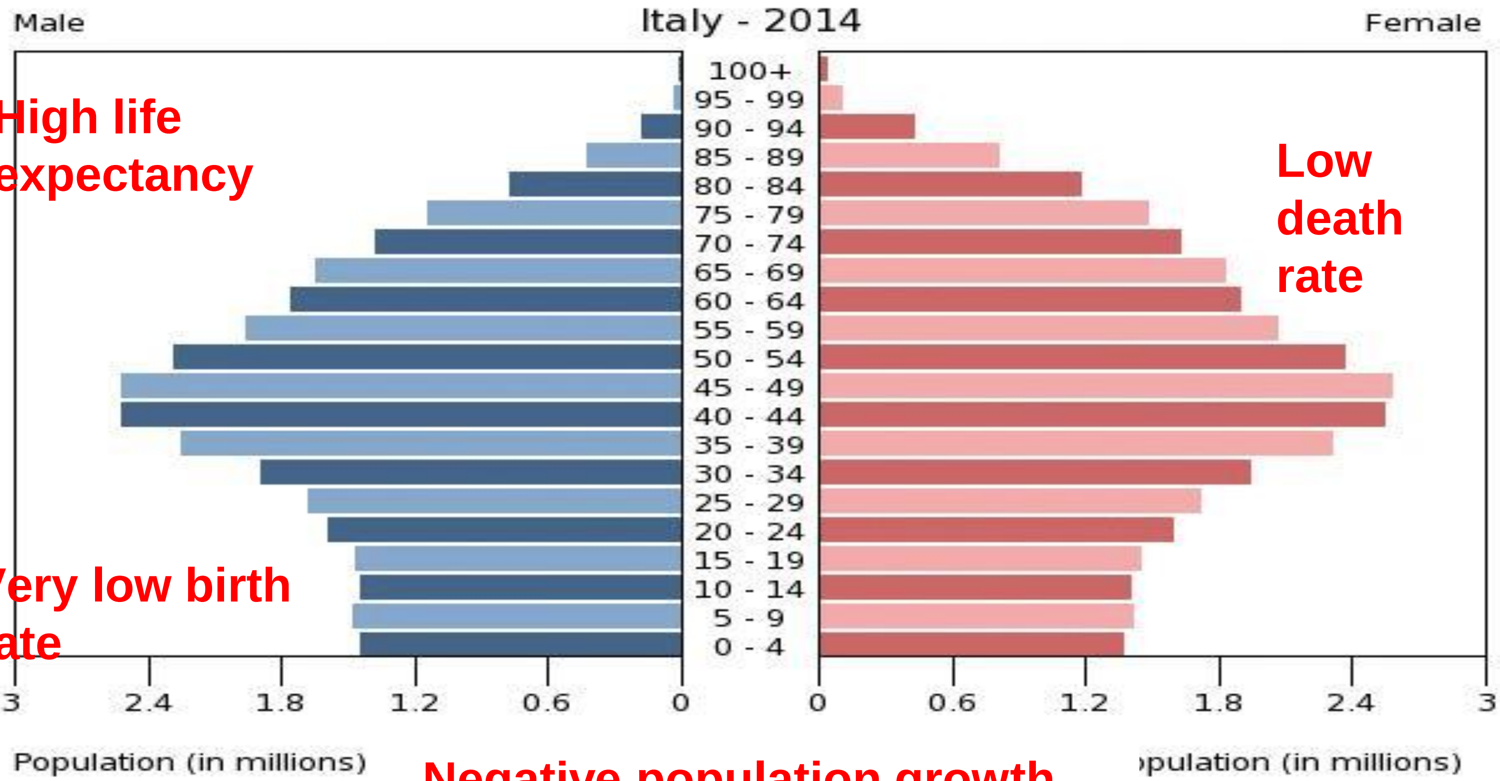
- Most preferable pyramid
- CBR, CDR, TFR Low
- TFR (~2.1) & IMR Declining/Low
  - $$ spent on education and jobs while also preparing for “graying nation”
- Slow pop. growth provides workforce that can support elderly population.
- MDCs (or NICs in transition)
- Youth dependency ratio low, but elder dependency ratio will begin to tick up
- Economy will become more service oriented
- Example: USA

Cup = Developed/Core (Negative Growth)

THE CUP!

High life expectancy

Low death rate



Very low birth rate

Negative population growth

Issues these countries are facing

- Negative Growth, shrinking population
- Low CBR, CDR, Low TFR (<2.1), low infant mortality
    - May develop programs to increase CBR, TFR or encourage immigration
- High Life Expectancy, MDCs
- Service Oriented Economy
- High old age dependency ratio: Gov't places more resources in retirement benefits "Graying nation;" strain on resources
- Example: Japan

What it means…

The **Triangle**
(Developing Country; rapid growth)

- High birth rate
- High or declining death rate
- Rapid population growth
- Shorter life expectancy

Like?
Angola
Cambodia

The **Box**
(More developed country; slow growth)

- Low Birth Rate/Death Rate
- Slow population growth
- Long life expectancy

Like?
Norway
USA

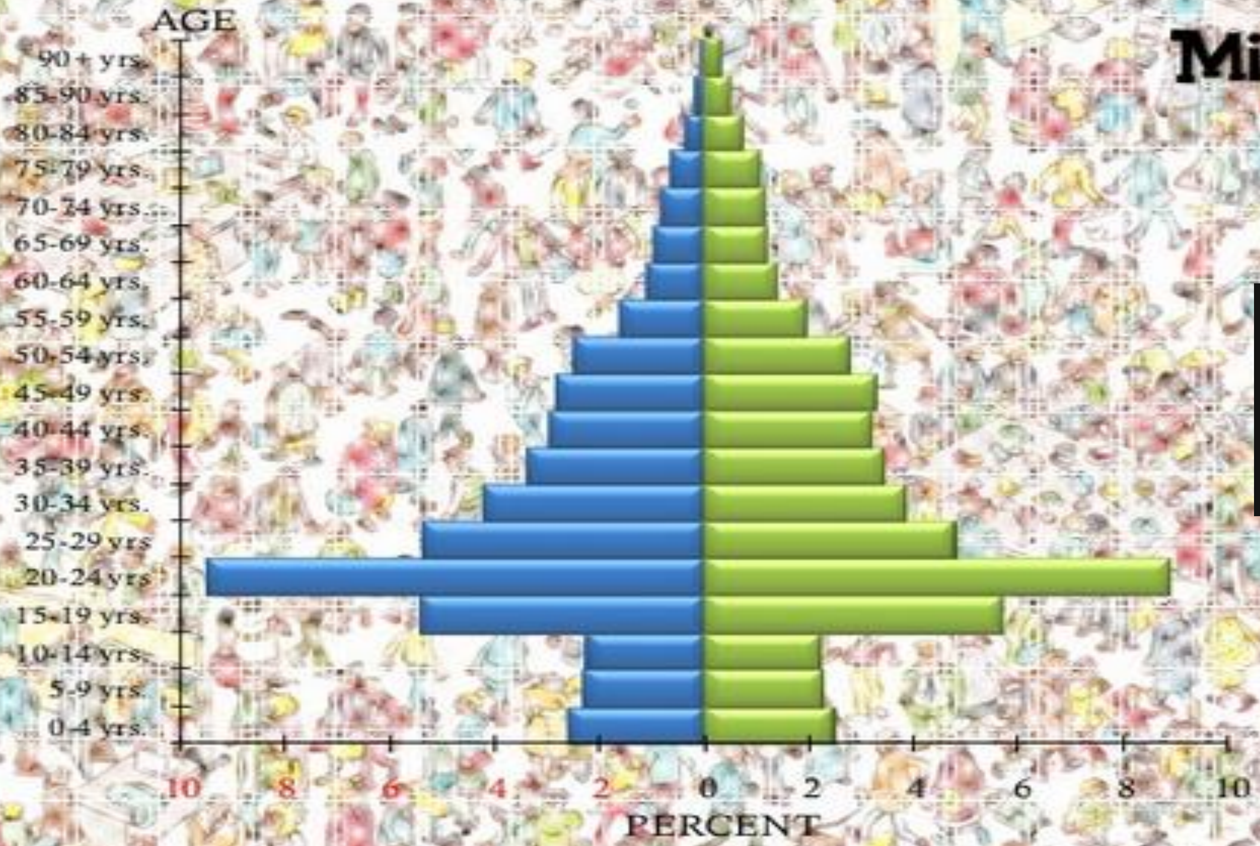
The **Cup**
(More developed country; negative growth)

- Very Low BR/Low DR
- Negative population growth
- Long life expectancy

Like?
Italy
Japan

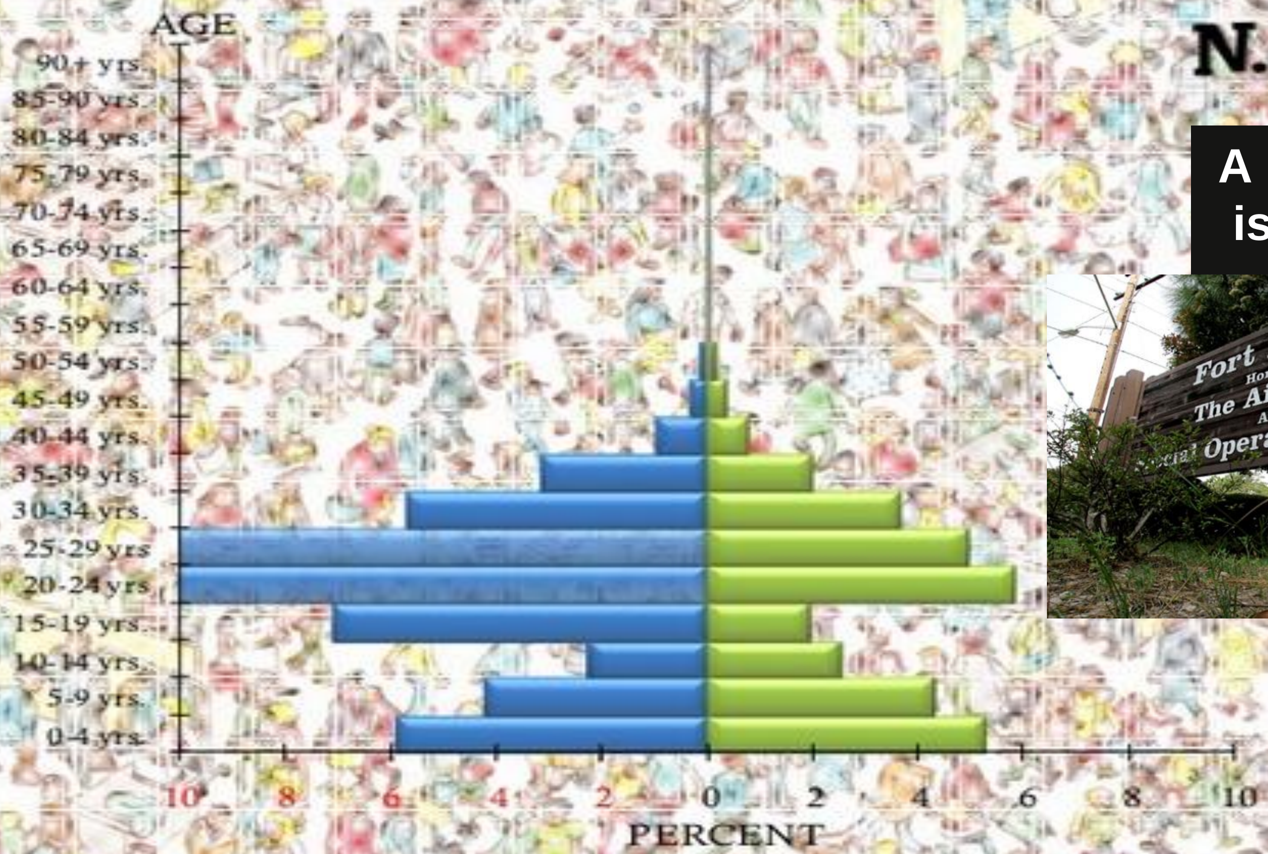
Pyramids with Anomalies

Ann Arbor, Michigan



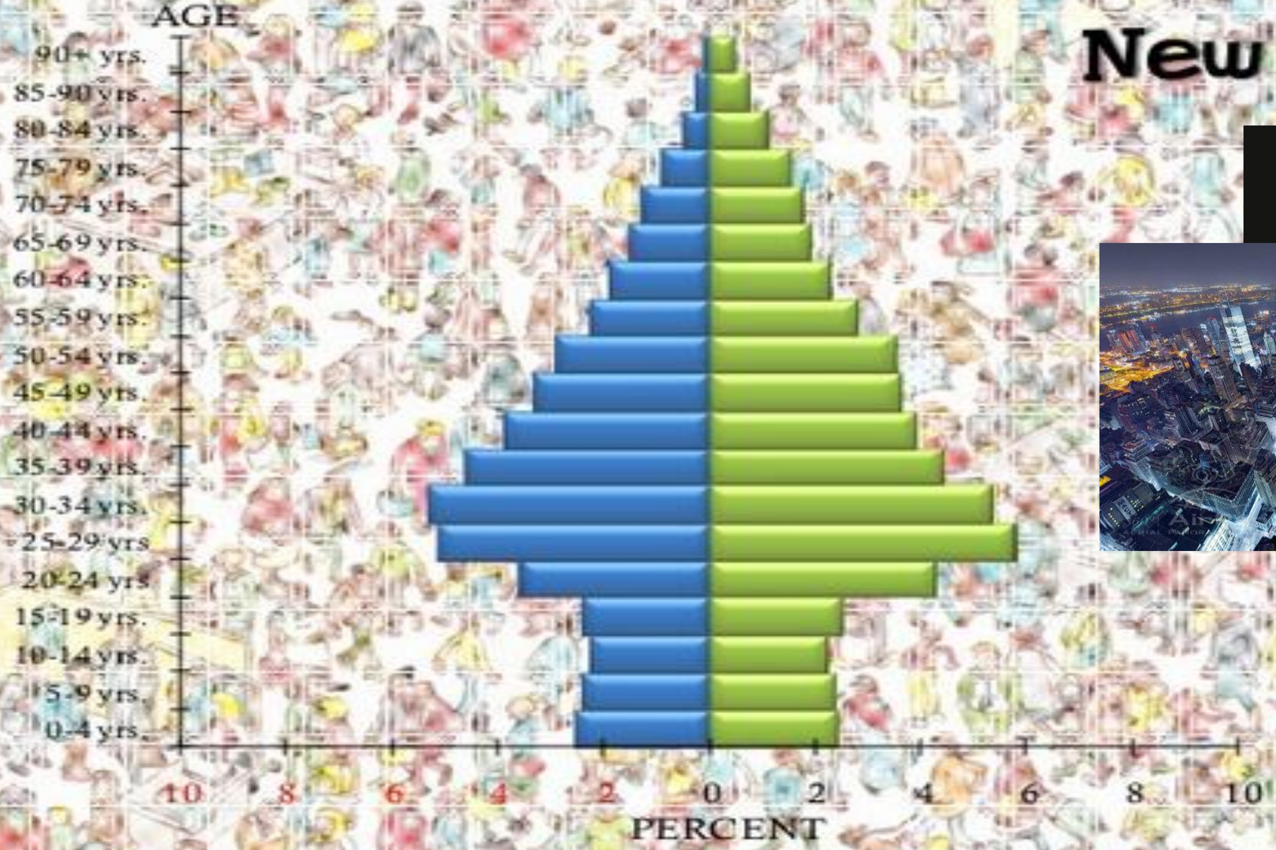
The University of Michigan is located here.

Ft. Bragg, N. Carolina



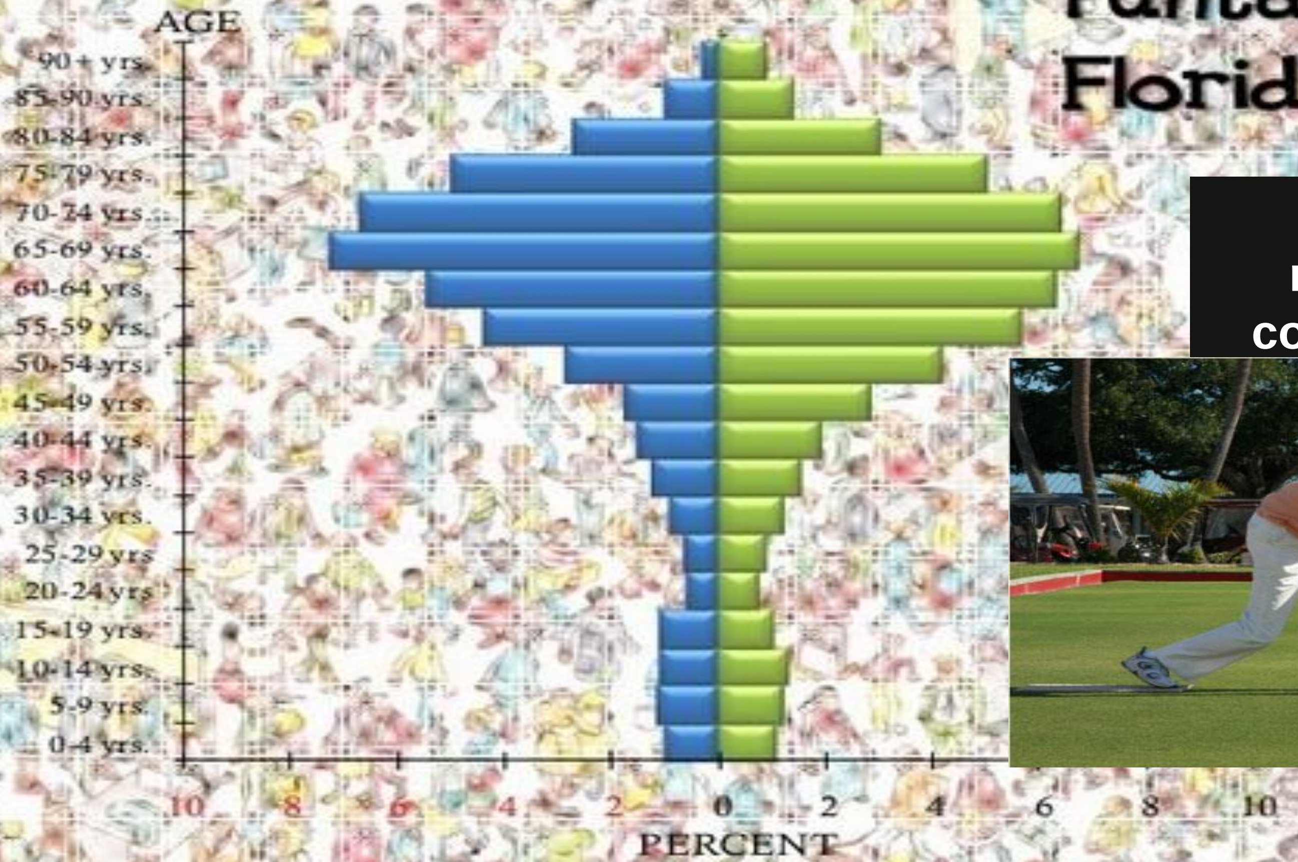
A US Army Base is located here.

Manhattan, New York



New York City

Punta Gorda, Florida



A large retirement community is

Northampton, Massachusetts

Smith College is located here.

Leavenworth, Kansas

A US Penitentiary is located here.

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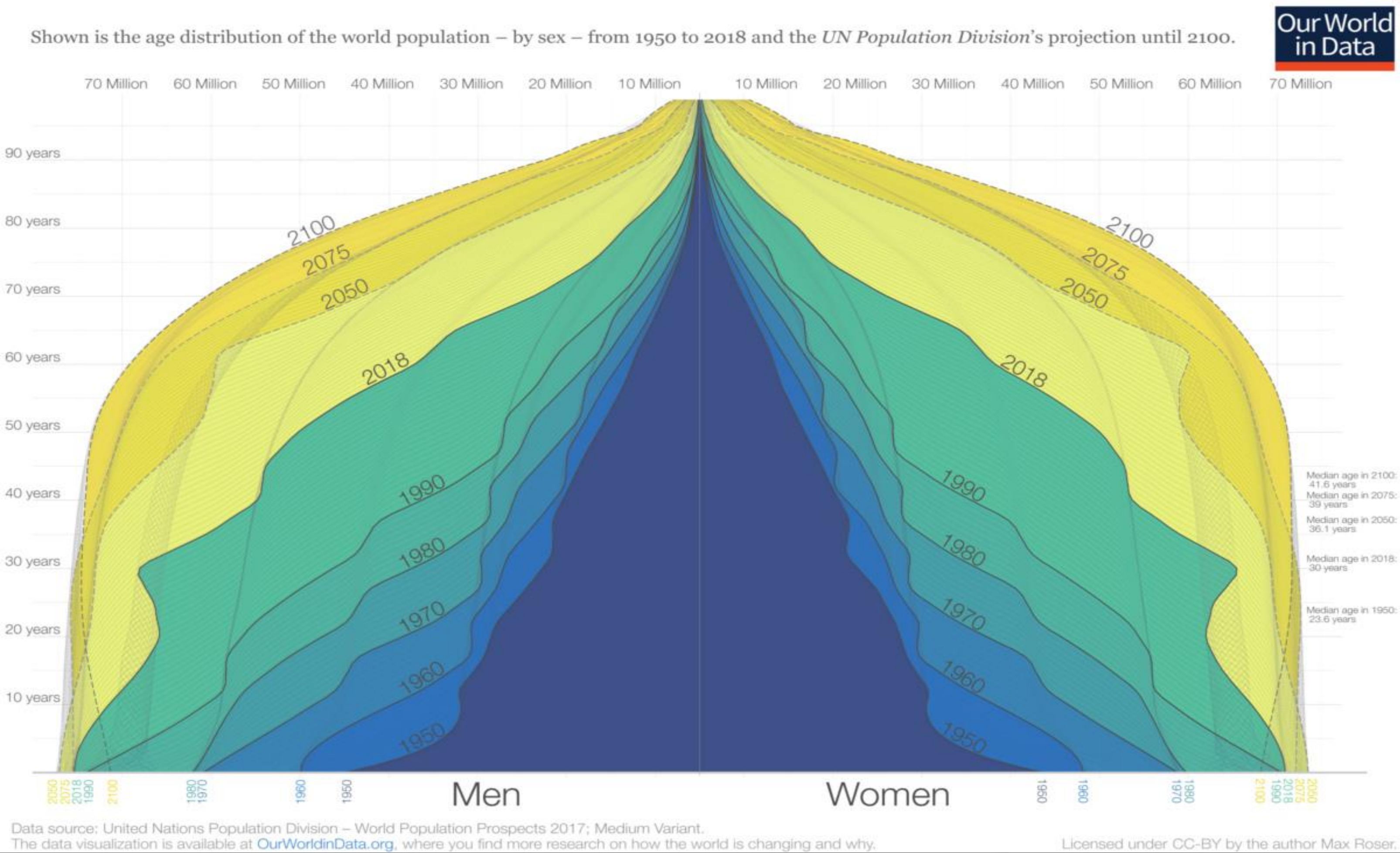
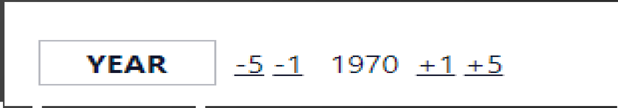


Population Pyramid Analysis: Go To: https://www.populationpyramid.net/world/2023/

1. Go to one of the countries you have already researched in Unit 1.5
2. Go to Year and add/subtract years as necessary to get pyramids for 1970, 2000 & 2030



1. Draw your pyramids and answer questions.